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Title:

SGLT2 inhibition reduces cardiac surgery-associated acute kidney injury: An open-label randomized study.

Short title:

MERCURI-1 trial

Authors:

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Abstract

Background

Cardiac surgery-associated acute kidney injury (CSA-AKI) is a common postoperative complication. Currently, no effective preventative strategies exist to mitigate CSA-AKI. Sodium-glucose transporter-2 (SGLT2) inhibitors reduced acute kidney injury (AKI) incidence in large, randomized placebo-controlled, cardiovascular and kidney outcome trials conducted in patients with chronic kidney disease. We hypothesized that perioperative SGLT2 inhibition could also reduce CSA-AKI.

Methods

In this open-label phase IV, randomized, parallel-group, pilot study, adult patients undergoing elective cardiac surgery with cardiopulmonary bypass were randomized to receive the SGLT2 inhibitor, empagliflozin (10 mg; oral), once daily three days prior to surgery and continued to two days after surgery compared with standard-of-care. Biomarkers for acute kidney injury (AKI), including serum and urinary neutrophil gelatinase-associated lipocalin (NGAL), serum and urinary kidney injury molecule-1 (KIM-1), and serum hypoxia-inducible factor-1α (HIF-1α) were measured. Additional outcomes included AKI incidence according to Kidney Disease: Improving Global Outcomes (KDIGO) criteria as well as metabolic parameters, including ketone body concentrations and glycemic control.

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Results

Between March 2022 and April 2023, 55 patients were included (sex: 73% male, age: 66 ± 10 years, BMI: 28 ± 4 kg/m^2, empagliflozin n = 25, control n = 30) in the intention-to-treat analysis. Empagliflozin significantly reduced the incidence of AKI (20% vs 66.7%; absolute difference 46.7%, 95% CI, -69.7 – -23.6; P=.001). Following surgery, urinary NGAL, and KIM-1 were found to increase in both arms, whereas a significant increment in serum HIF-1α after surgery was solely observed in the control group. We observed no between-group differences in the incidence of (euglycemic) ketoacidosis or hypoglycemic events.

Conclusions

Perioperative SGLT2 inhibition, compared with standard of care, significantly reduced the incidence of CSA-AKI. These findings warrant validation in large-scale, double-blind, placebo-controlled, randomized trials.

Trial Registry:

https://onderzoekmetmensen.nl/en/trial/26563 Identifier: NL9561

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Clinical perspective:

What Is New?

- In this open-label, randomized, controlled, pilot trial perioperative use of sodium glucose transporter-2 (SGLT2) inhibition with empagliflozin significantly reduced the incidence of acute kidney injury (AKI) by 46.7% (95% CI, -69.7 – -23.6; P=.001) compared to the control group.
- The level of ketone bodies increased significantly during cardiac surgery, however, there was no additional effect of empagliflozin treatment.

What Are the Clinical Implications?

- These results suggest that perioperative treatment with SGLT2 inhibitors might decrease the risk of cardiac surgery-associated (CSA)-AKI.
- These findings warrant validation in large-scale, double-blind, placebo-controlled, randomized trial, which is currently ongoing.

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88 **Non-standard Abbreviations and Acronyms**

89 AKI: Acute kidney injury

90 CI: Confidence interval

91 CKD: Chronic kidney disease

92 CPB: Cardiopulmonary bypass

93 CSA-AKI: Cardiac surgery-associated acute kidney injury

94 eGFR: estimated glomerular filtration rate

95 HIF1α: Hypoxia-inducible factor 1 alpha

96 IU: International units

97 KIM-1: Kidney injury molecule 1

98 NGAL: Neutrophil gelatinase-associated lipocalin

99 KDIGO: Kidney Disease: Improving Global Outcomes

100 SGLT2: Sodium-glucose transporter-2

101 T1D: Type 1 diabetes

102 T2D: Type 2 diabetes

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Introduction

Acute kidney injury (AKI) is a common complication following cardiac surgery, with a reported incidence of up to 70%. (1) Cardiac surgery-associated (CSA)-AKI is associated with postoperative morbidity and mortality, prolonged hospital stay, and increased healthcare costs. (2, 3) In addition, CSA-AKI can have long-term consequences, including the increased risk of developing chronic kidney disease (CKD) and reduced life expectancy compared to patients without CSA-AKI. (4-7)

Despite several attempts to prevent CSA-AKI with a variety of pharmacological interventions, no effective mitigation strategies have been developed to date. (2, 8, 9) Sodium-glucose transporter-2 (SGLT2) inhibitors have been reported to prevent AKI in large cardiovascular and kidney outcome trials in patients with or without type 2 diabetes (T2D). (10-13)

SGLT2 inhibitors reduce proximal sodium and glucose re-uptake, leading to an acute reduction in estimated glomerular filtration rate (eGFR) due to activation of tubuloglomerular feedback. This reduction in eGFR leads to a lesser kidney workload. Lower filtered sodium load can reduce kidney oxygen consumption, while SGLT2-induced improvements in insulin sensitivity and substrate metabolism may further reduce kidney oxidative requirements. (14-16) SGLT2 inhibitors may be protective in settings of renal tissue hypoxia during cardiac surgery, which has been directly reported in clinically relevant large animal models and in humans indirectly by measurement of bladder urinary oxygenation. (17, 18) However, the safety of SGLT2 inhibitors in cardiac surgery should also be considered, given the risk of SGLT2 inhibitor-associated perioperative ketoacidosis caused by the fasted state and surgical stress. (19) This serious side

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126 effect has prompted the Food and Drug Administration (FDA) to recommend interruption of
127 SGLT2 inhibitors at least 3 days prior to major surgery. (19, 20) Here, we performed an open-
128 label, randomized, pilot study to evaluate the effects of SGLT2 inhibition on biomarkers of AKI,
129 AKI incidence according to Kidney Disease: Improving Global Outcomes (KDIGO) criteria and
130 incidence of ketoacidosis and hypoglycemia in individuals undergoing cardiac surgery. We
131 hypothesized that perioperative SGLT2 inhibition with empagliflozin would reduce both
132 biomarkers of postoperative AKI and the incidence of AKI.

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Methods

Trial design

We performed a prospective, single-center, open-label, phase IV, randomized, controlled (standard care), balanced (1:1), pilot study. The study protocol was approved by the medical ethics committee of the Amsterdam UMC and by the Dutch competent authority before the initiation of the trial. The trial protocol was registered at the Dutch Trial Register (https://www.onderzoekmetmensen.nl), number NTR9561. An independent study monitor validated good clinical practice adherence and data collection quality. The study was conducted in compliance with the latest version of the Declaration of Helsinki and Good Clinical Practice guidelines. Neither a data safety management board nor a steering committee was involved in the decision-making of this study. All participants signed informed consent prior to any trial-related activity.

Study participants

Patients eligible for inclusion were aged between 18 and 90 years and scheduled to undergo elective cardiopulmonary bypass (CPB) assisted cardiac surgery at the Amsterdam University Medical Centre (UMC), a tertiary academic hospital in the Netherlands. Exclusion criteria included prior treatment with SGLT2 inhibitors, type 1 diabetes (T1D), T2D with a body mass index (BMI) below 25 (possibly phenotypically resembling T1D), estimated glomerular filtration rate (eGFR) <30 ml/min, systolic blood pressure below 100 mmHg, a history of ketoacidosis, a known or suspected allergy to the trial product, or individuals who were breastfeeding, intending to become pregnant or not using adequate contraceptive methods.

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Intervention

After providing written informed consent, participants were randomly assigned to the empagliflozin or control group. Patients in the empagliflozin group received a single dose of empagliflozin 10 mg orally (Boehringer Ingelheim International GmbH via Alliance Healthcare Nederland B.V.) every morning, starting from three days prior to surgery, until two days after surgery. The control group received no intervention and received standard perioperative care. We initially intended to start empagliflozin treatment seven days before surgery, however, due to logistical constraints in the planning of surgery, it became evident that this approach was not feasible. Therefore, we adjusted the number of treatments before surgery in the protocol from seven to three days prior to surgery before the recruitment of the first participant.

Procedures

Patients with T2D had their preoperative glucose management adjusted, if necessary, under the supervision of an endocrinology consultant, according to the treatment algorithm shown in Supplementary Figure 1. At the time of inclusion, on the morning of the day of surgery (before the first incision), and on the first and second morning after surgery (D1 and D2), serum and urine samples were collected to measure neutrophil gelatinase-associated lipocalin (NGAL), kidney injury molecule (KIM)-1, hypoxia-inducible factor 1 alpha (HIF1α) and creatinine. Furthermore, arterial blood gases and ketone levels were collected at four specific time points on the day of surgery: before the start of surgery (T1), at time of start of cardiopulmonary bypass (T2), at the end of cardiopulmonary bypass (T3) and at time of transport to the intensive care unit (ICU) (T4). As part of standard perioperative care, patients with insulin-dependent T2D

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177 received a continuous glucose/insulin infusion preoperatively. In case ketoacidosis was
178 suspected based on increased anion gap metabolic acidosis with the presence of ketone bodies,
179 patients were to be treated with a mixed glucose/insulin infusion to suppress ketogenesis and
180 subsequently closely monitored and treated according to local practice. Postoperatively,
181 creatinine levels and urinary output were measured according to usual care (creatinine was
182 measured daily and urinary output hourly until the removal of the urinary catheter upon
183 patients' discharge from ICU). We also recorded glucose levels as measured during routine care
184 (four times daily during ICU admission) and insulin administrations when prescribed by the
185 treating physicians. Adverse events were collected by reviewing in-hospital electronic health
186 records. The study workflow is summarized in Figure 1.

187 **Outcomes**

188 The primary outcome was the between-group difference of mean postoperative serum NGAL
189 concentrations measured on postoperative day two. Other outcomes included between-group
190 differences in any of the following measures: AKI incidence rates according to the KDIGO
191 criteria, postoperative urinary NGAL/creatinine ratios, serum KIM-1 concentrations, urinary
192 KIM-1/creatinine ratios, and serum HIF1α levels. Additionally, we calculated the difference in
193 biomarker concentrations between the first and second postoperative days compared to the
194 day of surgery. Metabolic secondary outcomes include the between-group difference of
195 intraoperative ketone levels, the number of ketoacidosis events (defined as a combination of
196 ketonemia >3 mmol/l in combination with a high-anion gap metabolic acidosis), the mean
197 intraoperative glucose level, the mean peak postoperative glucose level measured in the first
198 48 hours postoperatively, the incidence of hypoglycemia (defined as glucose < 4.0 mmol/L), the

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incidence of hyperglycemia (defined as glucose > 10.0 mmol/L) and the incidence of insulin administrations. Additionally, we compared the length of stay in the ICU and postoperative in-hospital complications such as mortality, renal replacement therapy, cerebrovascular accident (CVA), cardiac tamponade, gastrointestinal bleeding, anemia requiring transfusion, atrial fibrillation, other cardiac arrhythmia's, pacemaker implantations, infection of the mediastinum, pneumonia, and delirium.

ELISA

Serum and urinary concentrations of NGAL and KIM-1 were measured with, respectively, the LEGEND MAX™ Human NGAL (Lipocalin-2) ELISA Kit (Biolegend, cat.N. 443407) and the human KIM1 (TIM-1) ELISA Kit (ABCAM, cat.N. AB235081). Serum HIF-1A levels were detected using the HIF1A Human ELISA Kit (Invitrogen, cat.N. EHIF1A). All measurements were performed after the study had concluded and done according to the manufacturer's instructions. Urinary levels of NGAL and KIM-1 were normalized for concentrations of creatinine, detected by standard tests at the hospital laboratory of clinical chemistry.

Sample size

Based on a previous cohort study that found a difference in mean serum NGAL of 120 ng/ml between patients with and without AKI after cardiac surgery, we defined a between-group difference of 120 ng/ml in serum NGAL concentration on postoperative day 2 as a relevant outcome. (21) To detect a difference of 120 ng/ml, with 90% power, alpha at 0.05, we would need 40 patients per group.

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220 **Randomization and blinding**

221 Randomization was performed electronically through the data management system Castor EDC
222 (Ciwit BV, Amsterdam, the Netherlands) using balanced-block randomization, with variable-size
223 random computer-generated blocks of two, four, or six and an allocation ratio of 1:1. By design
224 (open-label study), patients, care providers nor study personnel were blinded to treatment
225 allocation.

226 **Statistical analyses**

227 We performed all analyses on the intention-to-treat population. When biomarkers were below
228 the detection limit, we assigned the value of zero. Discrete data are presented as counts with
229 percentages per group. Continuous variables are presented as mean ± SD or median
230 (interquartile range), depending on the distribution of the data. Within-group comparisons
231 were conducted using the paired Student t-test. Between-group comparisons were performed
232 using Fisher's exact test for dichotomous outcomes and unpaired Student's t-test or Mann-
233 Whitney U-tests for continuous variables, depending on the distribution of the data. Absolute
234 differences between groups are presented with their corresponding 95% confidence intervals
235 (95% CI). Repeated measurements were analyzed with generalized linear mixed models. To
236 compare between the treatment groups, we used the interaction term of the randomization
237 group with time. Additionally, we analyzed postoperative kidney biomarkers by calculating the
238 change in concentration compared to the day of surgery. Statistical analyses were performed
239 using SPSS (IBM version 24; IBM Corp., Armonk, New York).

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Results

Between March 2022 and March 2023, we enrolled 60 patients planned to undergo cardiac surgery, of which 29 patients were included in the empagliflozin group and 31 in the control group (Figure 2). Before the start of the intervention, five patients were excluded (one withdrew consent, two surgeries were cancelled for logistic reasons, and in two cases, the type of surgery was changed, thereby not meeting the inclusion criteria). All remaining patients were included in the intention-to-treat analysis (25 in the empagliflozin group, 30 in the control group). All patients in the intervention group started with empagliflozin three days prior to surgery.

Baseline characteristics of all included patients are summarized in Table 1. Overall, patient characteristics were well balanced between groups, except for age, with patients in the empagliflozin group being older. Patients were predominantly male, with relevant overweight, and had normal kidney function at baseline. T2D was present in 5 (9%) patients; none of them were treated with insulin therapy prior to hospital admission.

Kidney injury

The development of AKI characterized according to KDIGO criteria was observed in 5/25 (20%) patients in the empagliflozin group and 20/30 (66.7%) patients in the control group, with a difference of 46.7% between groups (95% CI, -69.7 – -23.6; P=.001) (Figure 3). In March 2023, the trial leadership team reviewed this significant between-group disparity in clinical outcomes. Concluding that these findings would require confirmation in a large, blinded and placebo-controlled trial, the trial leadership team ceased patient enrollment in this open-label pilot trial

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and notified the medical ethics committee of trial termination. Upon completion of data collection, blood and urine samples were sent for analysis.

Kidney biomarkers

Serum NGAL concentrations on the second postoperative day did not differ significantly between groups, empagliflozin: 70.8 ± 6.0 ng/ml vs control: 68.5 ± 10.0 ng/ml, with a difference of 2.3 ng/ml (95% CI, -6.9 – 2.3; P=.317). Other biomarkers for kidney injury, including urinary NGAL and KIM-1 in plasma and urine, did not demonstrate significant between-group differences either (Figure 4). In contrast, we did observe a difference between study groups in the development of serum HIF1α in the perioperative period. In the control group, HIF1α increased significantly from 110 ± 152 ng/ml before surgery to 171 ± 182 ng/ml on the first day after surgery (P=.04) and to 198 ± 268 ng/ml on the second day (P=.098). In contrast, in the empagliflozin group, HIF1α did not change from baseline levels in the postoperative period (Figure 4). A complete overview of all biomarker concentrations is available in Supplementary Table S1.

Metabolic effects

Ketone levels increased during surgery (Figure 5) in both groups; in 41 (76%) patients, the peak levels of ketones (1.1 ± 0.6 mmol/L) were measured at the end of CPB. The increase in ketone levels did not differ significantly between the groups. In the empagliflozin group, ketone levels increased by 1.0 ± 0.5 mmol/L compared to 0.8 ± 0.5 mmol/L in the control group, difference 0.2 mmol/L (95% CI, -0.5 – 0.1; P=.208). There was no significant difference in ketone

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development between the groups (P value of the interaction term time*group = .75). Ketoacidosis was not observed in any of the patients.

The mean intraoperative blood glucose was 6.6 ± 1.1 mmol/L in the empagliflozin group compared to 7.1 ± 1.5 mmol/L in the control group, difference 0.5 mmol/L (95% CI, -0.3 – 1.2; P=.194). The intraoperative incidence of hyperglycemia, hypoglycemia, and insulin administrations did not differ significantly between groups (Table 2). The peak glucose concentration during the first 48 hours postoperative was 9.3 ± 1.5 mmol/L in the empagliflozin group compared to 10.9 ± 2.5 mmol/L in the control group, difference of 1.6 mmol/L (95% CI, 0.5 – 2.8; P=.007). During the postoperative period, the incidence of observed hyperglycemia was 7/25 (28%) in the empagliflozin group compared to 18/30 (60%) in the control group, difference 32% (95% CI, -56.8 – -7.2; P=.029). In the empagliflozin group, 2/25 (8%) patients required insulin administrations due to hyperglycemia, compared to 11/30 (37%) patients in the control group, with a difference of 29% between groups (95% CI, 8.4 – 48.9; P=.023). None of the participants received insulin to treat ketoacidosis.

Adverse Events

No significant between-group differences were observed in the incidence of in-hospital postoperative complications (Supplemental Table 2). The length of stay in ICU was 23 (19-42) hours in the empagliflozin group compared to 32 (21-47) hours in the control group (P=.096).

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Discussion

In this open-label, randomized controlled, pilot study we investigated the effect of SGLT2 inhibition with empagliflozin on CSA-AKI. This pilot study was prematurely stopped due to the large between-group difference in AKI incidence; empagliflozin reduced the AKI incidence by 47% compared to the control group. HIF1α levels significantly increased postoperatively in the control group, whereas there was no significant change observed in the empagliflozin group. AKI reduction was not reflected in changes in NGAL and KIM-1. Ketone bodies increased significantly during surgery in both groups without an additional effect of empagliflozin. There was no significant between-group difference in the incidence of hypoglycemia (which occurred only once in the control group), while hyperglycemia was reduced.

CSA-AKI remains a frequent unresolved complication after cardiac surgery, with reported incidences varying between 20% and 70%, depending on the type of surgery, the definition of AKI used, and the protocol for AKI monitoring. (1) Our findings align with this high incidence, exemplified by an incidence of AKI in the control group of 66.7%, which can be attributed to our meticulous monitoring regimen of hourly urine output and daily creatinine measurements. This study provides compelling evidence that perioperative SGLT2 inhibition with empagliflozin offers protection against CSA-AKI. Our findings are in agreement with three meta-analyses consistently demonstrating a reduction in AKI incidence in large-scale cardiovascular and kidney outcome trials in outpatients. (10, 13, 22) The hypothesis that SGLT2 inhibitors provide kidney protection comes from the theory that SGLT2 inhibitors reduce kidney workload and may, therefore, improve kidney tissue oxygenation.

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To estimate the effect of empagliflozin on kidney hypoxia responses, we measured HIF1α levels, an established biomarker for hypoxia. We found that serum HIF1α levels remained stable in the empagliflozin group, whereas HIF1α increased in the control group after surgery. This is in line with earlier studies reporting that SGLT2 inhibition suppressed the HIF1-α pathway. (23, 24) Additionally, it has been shown (in mice) that SGLT2 inhibition after ischemia/reperfusion injury decreased the level of endothelial thinning, hypoxia, and the development of kidney fibrosis.(25)

While the clinical incidence of AKI was significantly lower after SGLT2 inhibition, this was not reflected by NGAL and KIM-1 differences. We selected serum and urinary NGAL and KIM-1 because they are widely studied biomarkers for AKI. (8) This study demonstrates that NGAL and KIM-1 do not reflect the kidney protective effects of SGLT2 inhibition following cardiac surgery.

In light of the FDA warning regarding the risk of ketoacidosis in patients perioperatively treated with SGLT2 inhibitors,(26) we measured the development of ketones during surgery. We observed a rise in ketone levels during surgery; however, no significant differences were observed between treatment groups. In addition, none of the patients were diagnosed with ketoacidosis during or after surgery. It is noteworthy that, by chance, insulin-dependent T2D patients were not included in this study. Had they been included, the risk of ketoacidosis would have been mitigated through the standard administration of glucose/insulin infusion perioperatively for insulin-dependent T2D patients. Although the limited sample size cannot prove safety, these findings are reassuring for further research. We also found no between-group differences in the incidence of hypoglycemia or other adverse events, while the risk for

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hyperglycemia was even reduced. The lack of increased hypoglycemia fits the data from trials in patients with T2D.(27)

By design, this pilot study tested for feasibility and was limited by its sample size with a standard care, open-label control group. As such, our findings warrant confirmation in a larger placebo-controlled trial. Our group is currently leading such a large multi-center trial that will include 784 patients undergoing cardiac surgery at multiple centers within the Netherlands and is estimated to complete enrollment in 2025. (clinicaltrials.gov: NCT05590143).

Conclusion

This pilot study indicates that SGLT2 inhibition may be a promising intervention for the reduction of CSA-AKI, requiring confirmation in a larger placebo-controlled double-blinded randomized controlled trial.

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Maartina Oosterom-Eijmael: Investigation, Formal Analysis, Validation, Writing - Original Draft.

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Yugeesh Lankadeva: Writing - Review & Editing, Supervision.

Mark Plummer: Writing - Review & Editing.

Benedikt Preckel: Resources, Writing - Review & Editing, Supervision.

Daniel van Raalte: Conceptualization, Methodology, Writing - Review & Editing, Supervision.

Jeroen Hermanides: Conceptualization, Methodology, Writing - Review & Editing, Supervision.

Abraham Hulst: Conceptualization, Methodology, Writing - Review & Editing, Supervision, Funding Acquisition.

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Disclosures

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464 **Tables**

465 **Table 1. Baseline characteristics of the intention-to-treat population.**

	All (n=55)	Empagliflozin (n=25)	Control (n=30)
Age, mean ± SD, years	66 ± 10	69 ± 8	63 ± 10
Male sex, n (%)	40 (73)	18 (72)	22 (73)
Ethnic origin, n (%)			
Caucasian	53 (96)	24 (96)	29 (97)
Other	2 (4)	1 (4)	1 (3)
BMI, mean ± SD, kg/m²	28.0 ± 4.4	27.6 ± 3.4	28.2 ± 5.1
T2D, n (%)	5 (9)	3 (12)	2 (7)
Diabetes medication, n (%)			
Metformin	4 (7)	3 (12)	1 (3)
Sulfonylurea derivate	1 (2)	0 (0)	1 (3)
Hypertension, n (%)	28 (51)	13 (52)	15 (50)
NYHA classification, n (%)			
I	18 (33)	10 (40)	8 (27)
II	28 (51)	12 (48)	16 (53)
III	9 (16)	3 (12)	6 (20)
LVEF, n (%)			
Good > 50%	40 (73)	20 (80)	20 (67)
Moderate 31-50%	15 (27)	5 (20)	10 (33)
ASA physical status classification, n (%)			
I	1 (2)	0 (0)	1 (3)
II	7 (13)	2 (8)	5 (17)

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III	45 (82)	22 (88)	23 (77)
IV	2 (4)	1 (4)	1 (3)
Creatinine clearance, mean ± SD, ml/min	70 ± 16	68 ± 16	73 ± 15
EuroSCORE II, mean ± SD, %	3.0 ± 3.2	2.8 ± 3.4	3.1 ± 3.0
Duration of surgery, mean ± SD, min	271 ± 96	249 ±83	291 ± 104
Type of surgery, n (%)			
CABG procedure	3(5.45)	1(4.0)	2 (6.67)
Single non-CABG procedure	28 (50.91)	12 (48)	16 (53.3)
Two or more procedures	24 (43.63)	12 (48)	12 (40)

ASA= American Society of Anesthesiologists; BMI= body mass index; CABG= coronary artery bypass graft; EuroSCORE= European System for Cardiac Operative Risk Evaluation; IQR= interquartile range; LVEF= left ventricular ejection fraction; NYHA= New York Heart Association; SD= standard deviation; T2D= type 2 diabetes

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468 **Table 2 Intraoperative and postoperative glycemic control and insulin administrations**

	Empagliflozin (n=25)	Control (n=30)	Absolute difference	95% CI	P value
Intraoperative					
Mean blood glucose, mean ± SD, mmol/L	6.6 ± 1.1	7.1 ± 1.5	0.5	-0.3 – 1.2	0.194
Hyperglycemia (>10 mmol/L), n (%)	3 (12)	6 (20)	-8	-27.2 – 11.2	0.487
Hypoglycemia (<4 mmol/L), n (%)	0 (0)	1 (3)	-3.3	-9.8 – 3.1	1.000
Insulin administration, n (%)	1 (4)	2 (7)	-2.7	-14.4 – 9.1	1.000
Postoperative[a]					
Peak blood glucose, mean ± SD, mmol/L	9.3 ± 1.5	10.9 ± 2.5	1.6	0.5 – 2.8	0.007
Hyperglycemia (>10 mmol/L), n (%)	7 (28)	18 (60)	-32	-56.8 – -7.2	0.029
Hypoglycemia (<4 mmol/L), n (%)	0 (0)	0 (0)	NA	NA	NA
Insulin administration, n (%)	2 (8)	11 (37)	-29	-48.9 – -8.4	0.023

CI= confidence interval; NA= not applicable; SD= standard deviation. [a] *The first 48 hours postoperative*

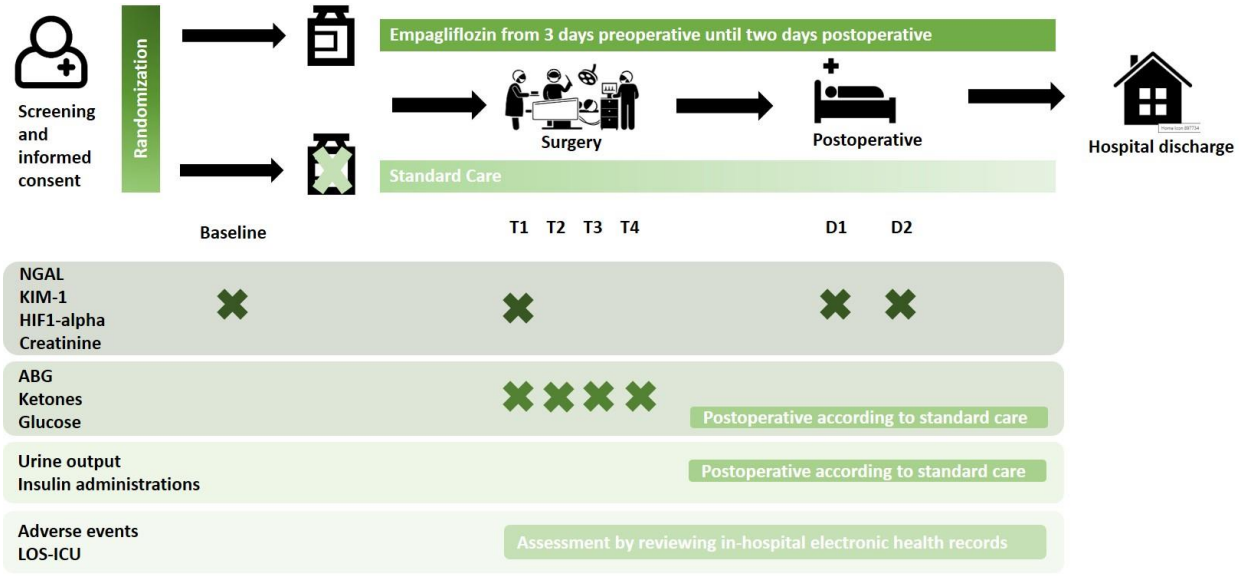
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Figures

Figure 1. Summary of study workflow.

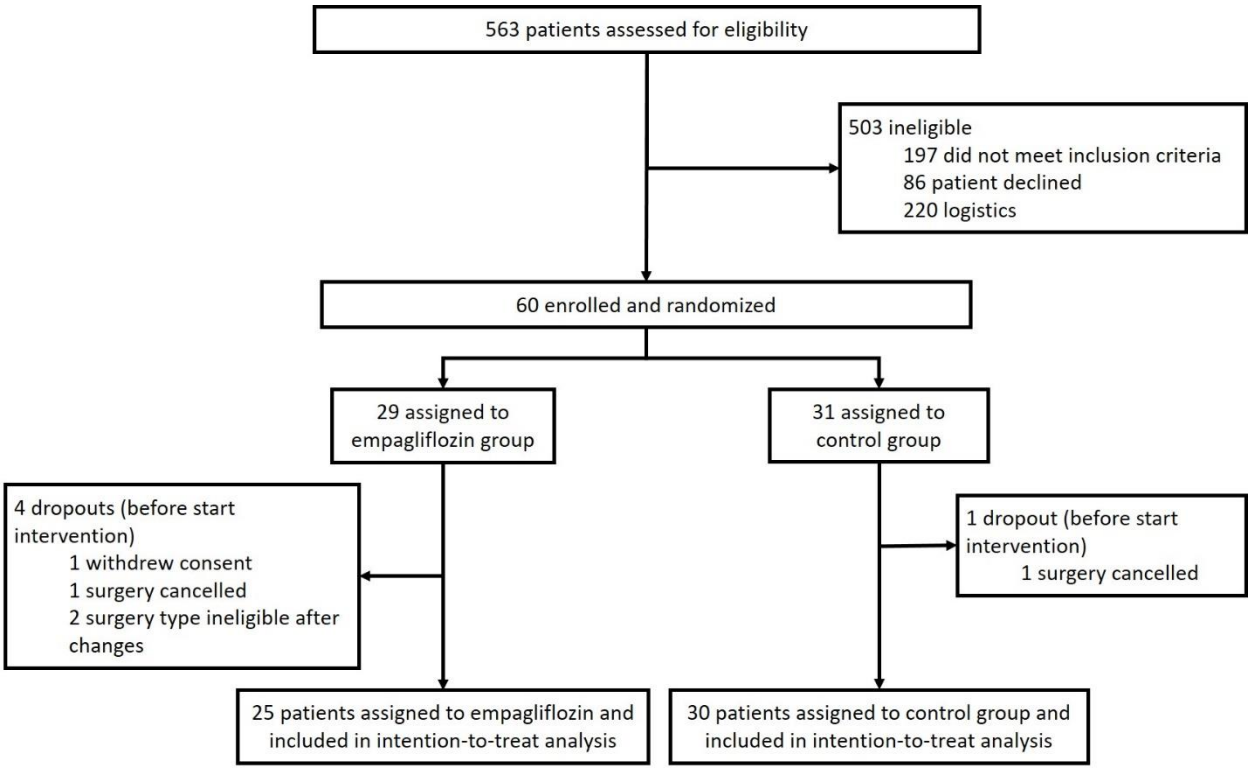


ABG= arterial blood gas; D1= first postoperative day; D2= second postoperative day; HIF1-alpha= Hypoxia-Inducible Factor1-Alpha; KIM-1= Kidney Injury Molecule-1; LOS-ICU= length of stay in the Intensive Care Unit; NGAL= Neutrophil Gelatinase-Associated Lipocalin; T1= before start of surgery; T2= at the start of cardiopulmonary bypass; T3= at the end of cardiopulmonary bypass; T4= at time of transport to the intensive care unit.

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478 **Figure 2. CONSORT flowchart of patients in the study**



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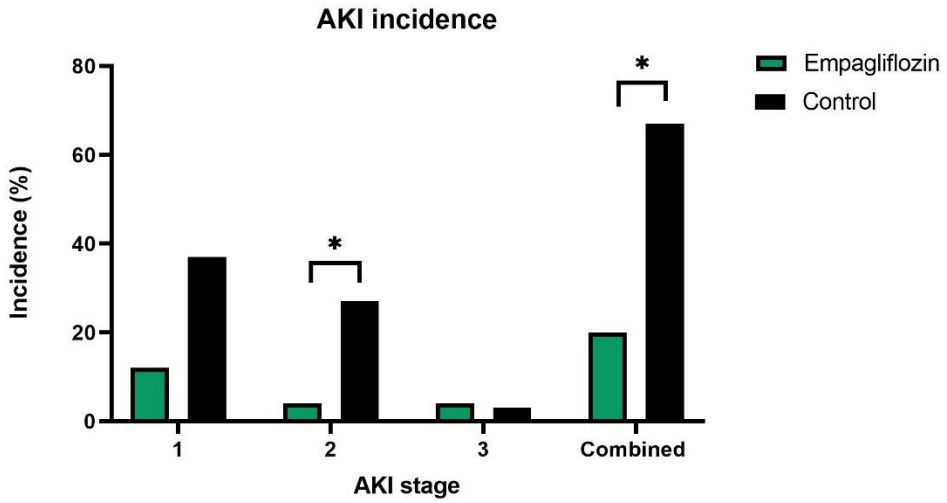
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485 **Figure 3. Incidence rates of Acute Kidney Injury (AKI) across various stages compared**

486 **between the empagliflozin and control group.**



	Empagliflozin	Control			
	25	30	Absolute difference	95% CI	P value
AKI, n (%)					
Stage 1	3 (12)	11 (36.7)	-24.7	-46.1 – -3.3	.061
Stage 2	1 (4)	8 (26.7)	-22.7	-40.3 – -5.1	.031
Stage 3	1 (4)	1 (3.3)	0.7	-9.3 – 10.7	1.000
Combined	5 (20)	20 (66.7)	-46.7	-69.7 – -23.6	.001

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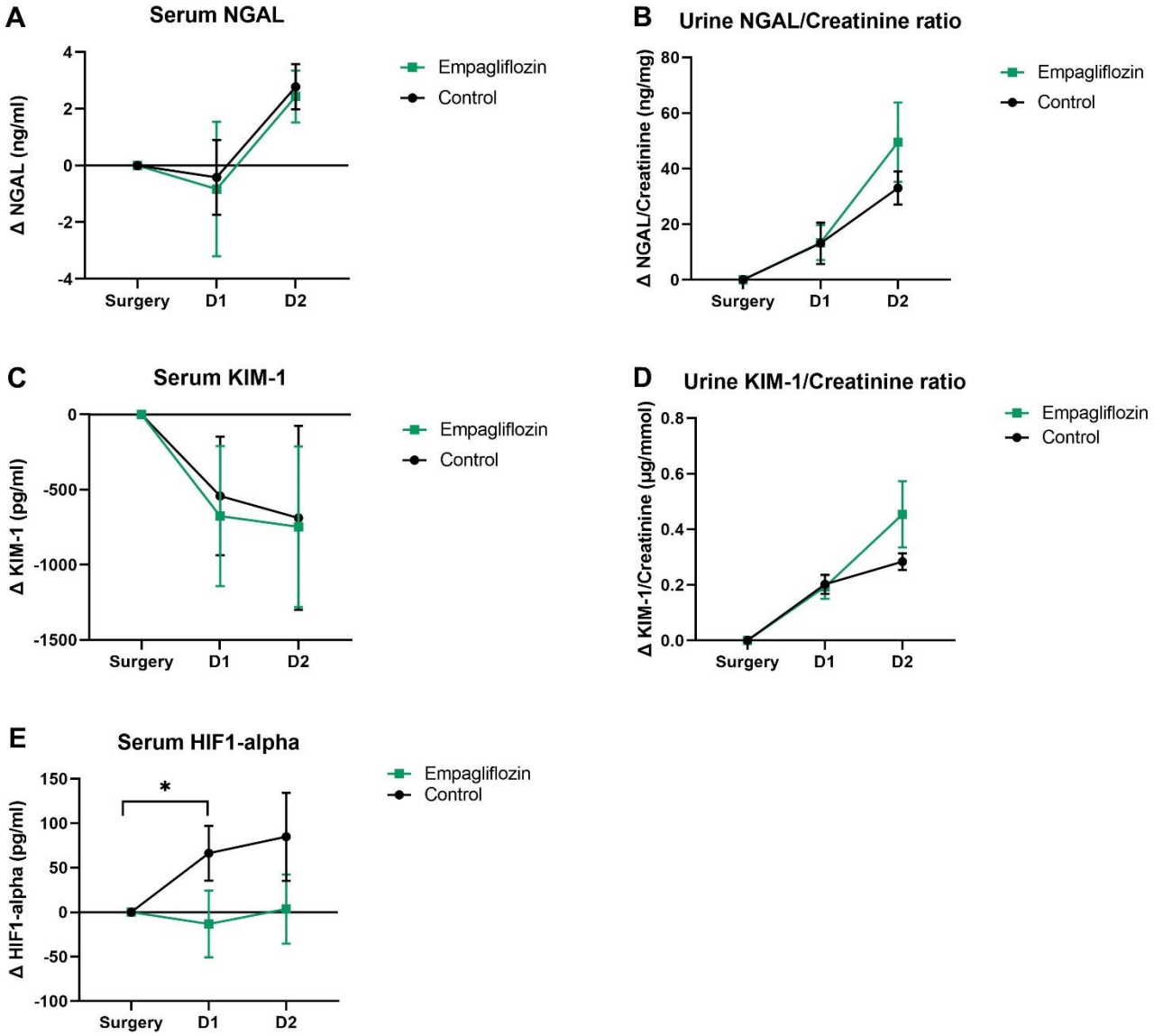
488 *AKI= acute kidney injury; CI= confidence interval*

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490 **Figure 4. Differences in postoperative kidney biomarkers concentrations compared to surgery**



491 **day for the empagliflozin and control group.**

492 *HIF1-alpha= Hypoxia-Inducible Factor1-Alpha; KIM-1= Kidney Injury Molecule-1; NGAL= Neutrophil Gelatinase-*

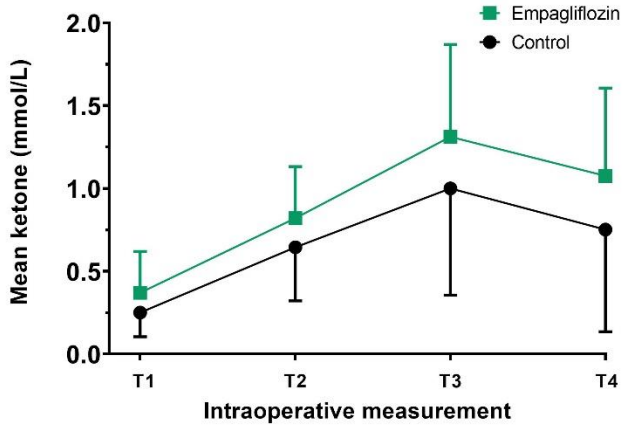
493 *Associated Lipocalin.*

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495 **Figure 5. Intraoperative (mean ± SD) ketone levels.**



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497 *T1= before start of surgery; T2= at the start of cardiopulmonary bypass; T3= at the end of cardiopulmonary*

498 *bypass; T4= at time of transport to the intensive care unit)*

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Supplementary Tables

Supplementary Table 1. Kidney biomarkers were measured on surgery day and the first two postoperative days.

		Empagliflozin (n=25)	Control (n=30)	Absolute Difference	95% CI	P value
Serum NGAL, (ng/ml)	**D0**	68.4 ± 6.0	66.3 ± 10.2	-2.1	-6.7 – 2.6	0.376
	D1	67.6 ± 11.4	65.4 ± 10.6	-2.2	-8.2 – 3.8	0.460
	D2	70.8 ± 6.0	68.5 ± 10.0	-2.3	-6.9 – 2.3	0.317
Urine NGAL/Creatinine (ng/mg)	**D0**	27.6 ± 58.2	9.0 ± 13.1	-18.6	-40.9 – 3.7	0.100
	D1	41.0 ± 74.0	21.8 ± 40.7	-19.2	-50.8 – 12.4	0.229
	D2	77.1 ± 120.8	41.2 ± 33.7	-35.9	-82.8 – 11.0	0.131
Serum KIM-1 (pg/ml)	**D0**	1458 ± 3843	1848 ± 4977	390	-2068 – 2849	0.751
	D1	841 ± 1694	1265 ± 3072	423	-636 – 1465	0.548
	D2	712 ± 1365	1126 ± 2300	414	-636 – 1465	0.432
Urine KIM-1/Creatinine (µg/mmol)	**D0**	0.2 ± 0.2	0.2 ± 0.1	0.0	-0.1 – 0.1	0.662
	D1	0.4 ± 0.3	0.4 ± 0.2	0.0	-0.1 – 0.2	0.893
	D2	0.7 ± 0.6	0.5 ± 0.2	-0.2	0.4 – 0.1	0.199
Serum HIF1α (pg/ml)	**D0**	142.7 ± 235.1	110.0 ± 151.6	-32.7	-139.3 – 73.9	0.541
	D1	129.5 ± 179.6	170.5 ± 182.5	41.0	-57.5 – 139.4	0.408
	D2	146.3 ± 134.7	198.1 ± 267.8	51.8	-66.5 – 170.1	0.384

Values are presented as mean ± SD. CI= confidence interval; D0= day of surgery; D1= first postoperative day; D2= second postoperative day; HIF1α= Hypoxia-Inducible Factor-1 alpha; KIM-1= Kidney Injury Molecule-1; NGAL= Neutrophil Gelatinase-Associated Lipocalin. SD= standard deviation

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Supplementary Table 2. Overview of in-hospital adverse events

	Empagliflozin (n=25)	Control (n=30)	P value
Mortality, n (%)	1 (4)	0 (0)	0.455
Renal replacement therapy, n (%)	1 (4)	1 (3)	1.000
CVA, n (%)	1 (4)	1 (3)	1.000
Cardiac tamponade, n (%)	3 (12)	4 (13)	1.000
Pericardial effusion, n (%)	0 (0)	1 (3)	1.000
Thromboembolic events, n (%)	1 (4)	0 (0)	0.455
Gastrointestinal bleeding, n (%)	1 (4)	0 (0)	0.455
Anemia requiring transfusion, n (%)	2(8)	3 (10)	1.000
Atrial fibrillation, n (%)	6 (24)	7 (23)	1.000
Other cardiac arrhythmia, n (%)	4 (16)	3 (10)	0.689
Pacemaker implementation, n (%)	2 (8)	1 (3)	0.585
Infection mediastinum, n (%)	2 (8)	0 (0)	0.202
Urinary tract infection, n (%)	1 (4)	0 (0)	0.455
Pneumonia, n (%)	1 (4)	0 (0)	0.455
Delirium, n (%)	0 (0)	1 (3)	1.000

Abbreviations: CVA, Cerebrovascular accident

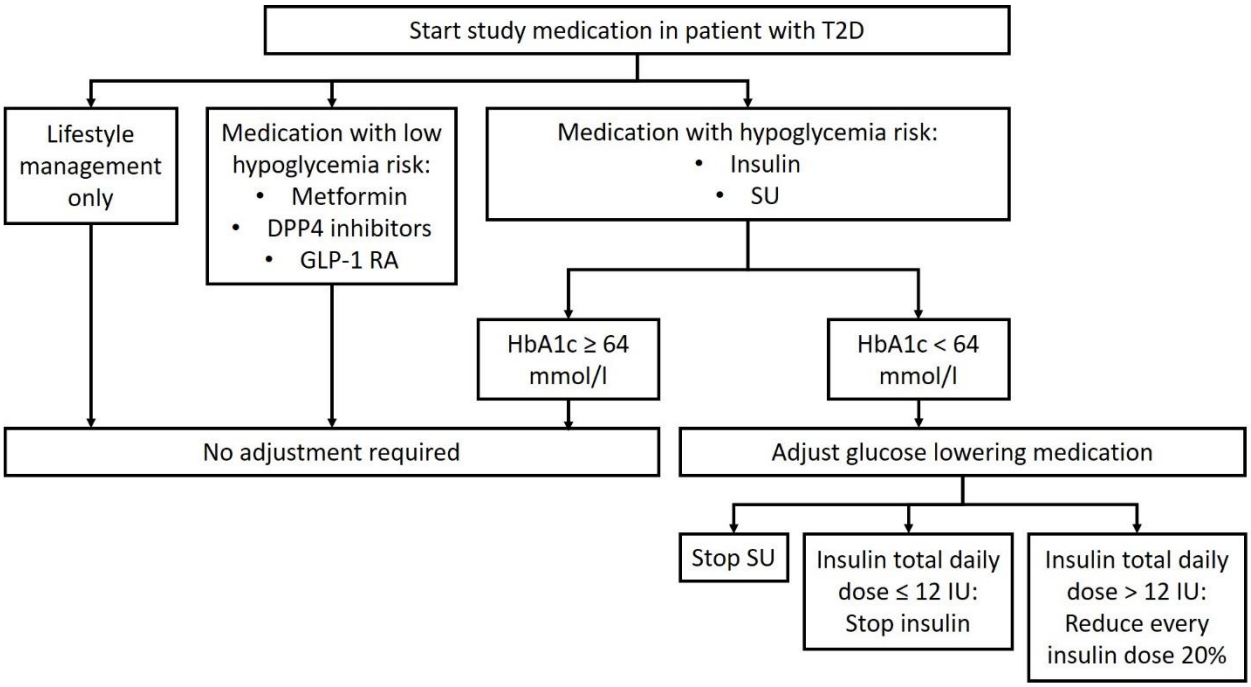
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523 ***Supplemental Figures and Figure Legends***

524 **Supplemental Figure 1. Treatment algorithm for adjustment of antihyperglycemic therapy in**

525 **patients with a known history of type 2 diabetes**



526

527 *DPP4 inhibitors= Dipeptidyl Peptidase-4 inhibitors; GLP-1 RA= Glucagon-Like Peptide-1 Receptor Agonist; IU=*

528 *international units; SU= Sulfonylurea; T2D= Type 2 Diabetes*

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